COMPARISON TESTS

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Name: Solutions

Theorem (The Comparison Tests)**.** *Let $\{a_n\}$ and $\{b_n\}$ be sequences, and assume there exists some number N such that*

$$0 < a_n \leq b_n$$

is satisfied whenever $n \geq N$.

(i) If $\sum a_n$ diverges, then $\sum b_n$ also diverges.

(ii) If $\sum b_n$ converges, then $\sum a_n$ also converges.

Theorem (The Limit Comparison Test)**.** *Let $\{a_n\}$ and $\{b_n\}$ be sequences, and assume there exists some number N such that*

$$0 < a_n, b_n$$

is satisfied whenever $n \geq N$. If there exists some number $c > 0$ such that

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$$

then either

- *$\sum a_n$ and $\sum b_n$ both converge, or*
- *$\sum a_n$ and $\sum b_n$ both diverge.*

Decide whether the following series converge or diverge.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{9^n}{3+10^n}$ Converges by Comparison:

$$10^n \leq 3+10^n \Rightarrow \frac{1}{3+10^n} \leq \frac{1}{10^n} \Rightarrow \frac{9^n}{3+10^n} \leq \frac{9^n}{10^n}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{9^n}{3+10^n} \leq \sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n = \sum_{n=1}^{\infty} \frac{9}{10}\left(\frac{9}{10}\right)^{n-1} = \frac{\frac{9}{10}}{1-9/10} = \frac{9}{10(1-9/10)} = \frac{9}{10-9} = 9$$

2. $\sum_{k=1}^{\infty} \frac{(2k-1)(k^2-1)}{(k+1)(k^2+4)^2}$ Converges by L.C.T with $\sum_{k=1}^{\infty} \frac{2}{k^2}$

$$\lim_{k\to\infty} \frac{(2k-1)(k^2-1)}{(k+1)(k^2+4)^2}\left(\frac{k^2}{2}\right) = \lim_{k\to\infty} \frac{(2k-1)(k-1)k^2}{2(k^2+4)^2}$$

$$= \lim_{k\to\infty} \frac{2k^4-2k^3-k^3+1}{2k^4+16k^2+32} = 1 > 0$$

3. $\sum_{n=1}^{\infty} \frac{1+\cos(n)}{e^n}$

Converges by Comparison

$$1+\cos(n) \le 1+1=2 \Rightarrow \frac{1+\cos(n)}{e^n} \le \frac{2}{e^n} = \frac{2}{e}\left(\frac{1}{e}\right)^{n-1}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1+\cos(n)}{e^n} \le \sum_{n=1}^{\infty} \frac{2}{e}\left(\frac{1}{e}\right)^{n-1} = \frac{2}{e}\left(\frac{1}{1-1/e}\right) = \frac{2}{e-1}$$

4. $\sum_{n=1}^{\infty} \frac{2}{\sqrt{n}+2}$ Diverges by L.C.T with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

$$\lim_{n\to\infty} \frac{2}{\sqrt{n}+2}\left(\frac{\sqrt{n}}{1}\right) = \lim_{n\to\infty} \frac{2\sqrt{n}}{\sqrt{n}+2} = 2 > 0.$$

5. $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$

Diverges by L.C.T. with the Harmonic Series:

$$\lim_{n\to\infty} \frac{\sin(\frac{1}{n})}{\frac{1}{n}} = \lim_{x\to 0} \frac{\sin(x)}{x} = 1 > 0.$$